

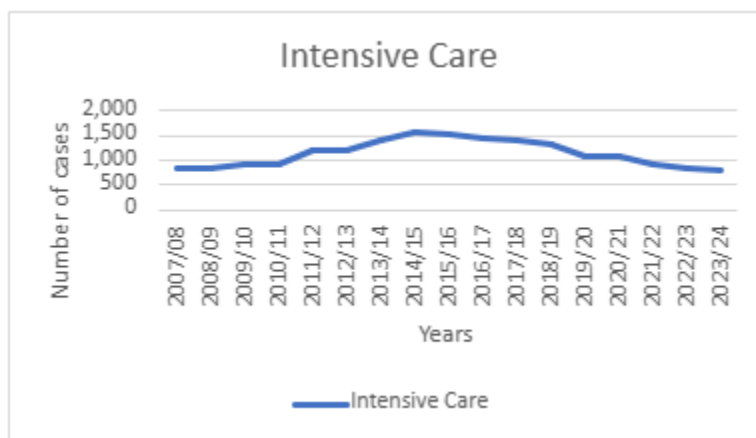
Part A: Forecasting Business Analytics

Question 1

This report presents an analysis of the time series data for three levels of neonatal critical care: **Intensive Care**, **High Dependency Care**, and **Special Care**. The objective is to visualise the trends over time and identify any key time series characteristics such as trends, seasonality, or fluctuations.

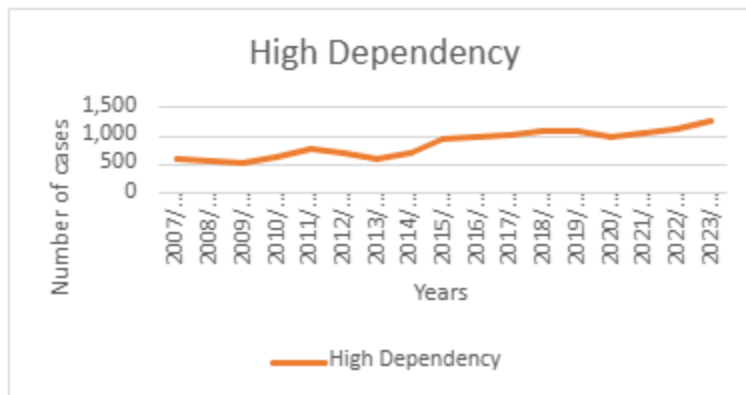
Time Series Plots

Below are the three separate time series graphs for each level of care, presenting the number of neonates receiving care over the years:



1. Intensive Care (IC)

The graph for **Intensive Care** shows a generally rising trend from 2007 until roughly 2015, followed by a modest dip in subsequent years. This pattern might reflect earlier breakthroughs in medical treatment, which resulted in more newborns requiring critical care. The decline after 2016 might indicate better prenatal care, lessening the need for essential intervention at delivery. While there are some minor shifts over time, there does not appear to be a distinct seasonal pattern.



2. High Dependency Care (HDC)

The graph for **High Dependency Care** has some ups and downs, but the main trend is upward. There was a considerable dip between 2013 and 2014, however since 2015, the numbers have been progressively increasing. This rise might be attributed to more newborns surviving early problems, which means they often require additional care until they fully recover. The swings might also indicate changes in hospital regulations or an increasing population, both of which could influence the need for **High Dependency Care**.



3. Special Care (SC)

The time series for **Special Care** shows a steady decline from 2007 to around 2020, with only slight changes in between. A particularly sharp drop appears between 2019 and 2021, possibly due to updated healthcare practices or improvements in infant care that reduce the need for longer-term special support. This decrease might indicate improved, more efficient neonatal care, reducing the need for less intense therapies. Unlike **Intensive** and **High Dependency Care**, **Special Care** has not had the same long-term increase, which might indicate that its function in infant care is evolving in a different way.

Conclusion

The time series study of newborn critical care levels gives useful information about patterns and healthcare demand across time. While Intensive Care originally climbed before somewhat falling, High Dependency Care has steadily grown with variations, while Special Care has consistently declined. These findings can assist policy makers and healthcare planners better allocate resources to meet future newborn care demands.

Question 2

2A) Naive Method

C3								
4755								
	A	B	C	D	E	F	G	H
1	Year	Special Care	Naive Method Forecast	Forecast Error	Squared Forecast Error		Absolute Error Forecast	
2	2007/08	4,755						
3	2008/09	4,758	4,755	3	9		3	
4	2009/10	4,596	4,758	-162	26,244		162	
5	2010/11	3,992	4,596	-604	364,816		604	
6	2011/12	4,763	3,992	771	594,441		771	
7	2012/13	4,525	4,763	-238	56,644		238	
8	2013/14	3,945	4,525	-580	336,400		580	
9	2014/15	3,652	3,945	-293	85,849		293	
10	2015/16	3,944	3,652	292	85,264		292	
11	2016/17	3,619	3,944	-325	105,625		325	
12	2017/18	3,359	3,619	-260	67,600		260	
13	2018/19	3,132	3,359	-227	51,529		227	
14	2019/20	2854	3,132	-278	77,284		278	
15	2020/21	2095	2,854	-759	576,081		759	
16	2021/22	2316	2095	221	48,841		221	
17	2022/23	1985	2316	-331	109,561		331	
18	2023/24	2340	1985	355	126,025		355	
19				MSE=	173413.2857	MAD=	356	MAPE=
								11

This table implements the Naïve forecasting technique, with the formula $F_{t+1} = A_t$. F_{t+1} represents the forecast for the next period, while A_t represents the actual value from the current period. Each year's forecast merely replicates the previous year's actual result, making this approach simple and easy to implement. The forecast error, squared error, absolute error, and absolute percentage error (APE) are calculated on an annual basis to measure accuracy. Key performance measures such as Mean Squared Error (MSE), Mean Absolute Deviation (MAD), and Mean Absolute Percentage Error (MAPE) are then summed at the bottom of the table for a thorough examination. While this approach is not appropriate for datasets with obvious patterns or seasonality, a MAPE of 11% suggests that it delivers relatively accurate short-term forecasts, especially when the data is steady and consistent.

Business Analytics

2B) 4 Period Moving Average

C26 X fx =AVERAGE(B22:B25)									
	A	B	C	D	E	F	G	H	I
20									
21	Year	Special Care	4 Period MA Forecast	Forecast Error	Squared Forecast Error		Absolute Error Forecast		APE (%) 4-Period MA
22	2007/08	4,755							
23	2008/09	4,758							
24	2009/10	4,596							
25	2010/11	3,992							
26	2011/12	4,763	4,525	238	56,525		238		4.991601932
27	2012/13	4,525	4,527	-2	4		2		0.044198895
28	2013/14	3,945	4,469	-524	274,576		524		13.28263625
29	2014/15	3,652	4,306	-654	427,716		654		17.90799562
30	2015/16	3,944	4,221	-277	76,729		277		7.023326572
31	2016/17	3,619	4,017	-398	158,404		398		10.99751313
32	2017/18	3,359	3,790	-431	185,761		431		12.83119976
33	2018/19	3,132	3,644	-512	262,144		512		16.34738186
34	2019/20	2854	3514	-660	435,600		660		23.12543798
35	2020/21	2095	3241	-1,146	1,313,316		1,146		54.70167064
36	2021/22	2316	2860	-544	295,936		544		23.48877375
37	2022/23	1985	2599	-614	376,996		614		30.93198992
38	2023/24	2340	2313	27	729		27		1.153846154
39			MSE=	297264.3125	MAD=	463.5961538	MAPE=	16.67904404	

The figure above forecasts future values using a 4-period moving average that averages the preceding four years. This was calculated by using the formula: $F_t = (A_{t-1} + A_{t-2} + A_{t-3} + A_{t-4}) / 4$. This method smoothes out short-term variations and works well with underlying patterns, although it can be slow to react to rapid changes. The table also displays the forecast errors and critical performance parameters such as MSE, MAD, and MAPE. The MAPE of 16.68% is somewhat less accurate than the naïve technique, most likely because of considerable year-to-year changes in the data.

2C) Exponential Smoothing method

C43 X fx =(0.2*B43)+(0.73*C42)											
	A	B	C	D	E	F	G	H	I	J	K
41	Year	Special Care	ES Forecast	Forecast Error	Squared Forecast Error		Absolute Error Forecast		APE (%) ES		Forecast Error/Actual
42	2007/08	4,755	4,755								Alpha (a) = 0.73
43	2008/09	4,758	4,423	335	112,393		335		7		0
44	2009/10	4,596	4,192	404	163,188		404		9		0
45	2010/11	3,992	3,901	91	8,371		91		2		0
46	2011/12	4,763	3,839	924	853,823		924		19		0
47	2012/13	4,525	3,746	779	607,089		779		17		0
48	2013/14	3,945	3,561	384	147,516		384		10		0
49	2014/15	3,652	3,365	287	82,092		287		8		0
50	2015/16	3,944	3,279	665	441,883		665		17		0
51	2016/17	3,619	3,150	469	219,539		469		13		0
52	2017/18	3,359	3,003	356	126,641		356		11		0
53	2018/19	3,132	2,849	283	80,248		283		9		0
54	2019/20	2854	2,679	175	30,677		175		6		0
55	2020/21	2095	2,401	-306	93,850		-306		15		0
56	2021/22	2316	2,240	76	5,746		76		3		0
57	2022/23	1985	2,055	-70	4,865		-70		4		0
58	2023/24	2340	1,989	351	123,543		351		15		0
59			MSE=	193841.4949	MAD=	325.2042508	MAPE=	10.27098638			

The Exponential Smoothing approach, applying the formula $F_t = \alpha \times A_{t-1} + (1 - \alpha) \times F_{t-1}$, was used to create the forecasts shown in this final graph. In this equation, F_t represents the current forecasting, A_{t-1} is the previous actual value, F_{t-1} is the last forecast, and α (0.73) is the smoothing factor. Because this strategy prioritises current data, it responds faster to changes than the moving average. It also produced overall the best results, with a MAPE of just 10.27%—the most accurate of the three approaches. These findings suggest that Exponential Smoothing is the most effective method for predicting special care admissions in this dataset.

2D)

Compare all three Forecasting methods			
Methods	MSE	MAD	MAPE
Naive method	173,413	356	11
4 Period MA	297,273	464	17
ES	193,841	325	10

The table compares three forecasting methods based on MSE, MAD, and MAPE to evaluate accuracy and error magnitude. Exponential Smoothing (ES) is the most accurate, with the lowest MAPE (10%) and lowest MAD (325), making it the best choice. The Naïve Method performs well and shows high accuracy, with a MAPE of (11%) and the lowest MSE (173,413). The 4-period Moving Average is the least accurate, with the largest MAPE (17%) and MAD (464), making it unsuitable for predicting.

Conclusion

Exponential Smoothing (ES) is the best approach due to low mistake rates, followed by the **Naïve** approach. **Four-period MA** is not recommended owing to its high error rate.

Question 3

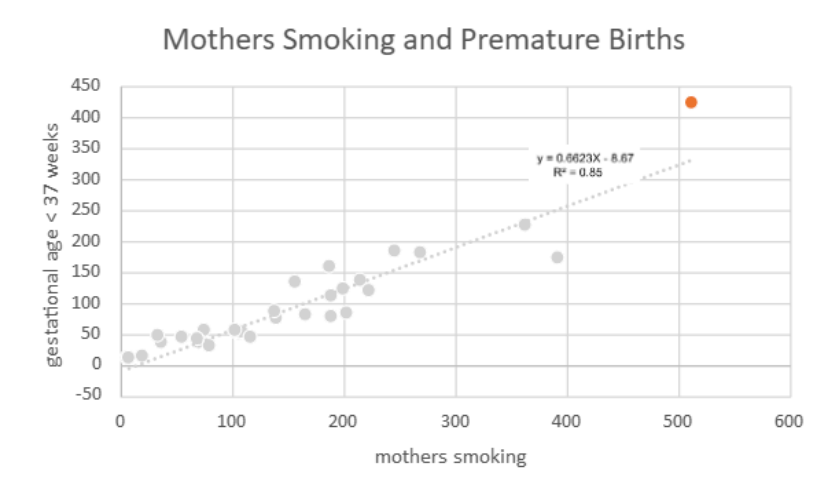
Recommendation of Forecasting Technique for Special Care Neonates

Based on the results of Question 2, **Exponential Smoothing (ES)** is the most successful forecasting approach. It had a low Mean Absolute Percentage Error (MAPE) of 10% and the least Mean Absolute Deviation (MAD) of 325, indicating high accuracy and consistency. Unlike the **Naïve method** approach and the **4-period Moving Average**, **ES** adjusts to current trends by focussing on the most recent data, reducing forecast errors. From a theoretical approach, exponential smoothing is frequently recommended for time series forecasting in fields such as demand planning and healthcare (Makridakis et al., 2018). It works especially effectively when data exhibits short-term volatility but follows a broad pattern, such as changing demand for newborn care. To further improve accuracy, an optimised smoothing constant (α) should be selected to minimize forecasting errors. Additionally, advanced methods like Holt's Exponential Smoothing (which accounts for trends) or ARIMA models could be considered for long-term forecasting, but they require more complex parameter tuning. To better understand and plan for the demand on **Special Care (SC)** neonatal facilities, we've developed a forecasting model using **Exponential Smoothing**. The model enhances future prediction accuracy by evaluating previous data and employing an optimised smoothing factor. This allows healthcare planners to make better informed decisions about staffing levels and the availability of neonatal facilities.

Conclusion:

Exponential Smoothing is the best method to go for as it's a valuable and reliable method for predicting. Its versatility and simplicity make it ideal for healthcare planning, where timely and precise projections may make a significant impact on patient care.

Question 4



Analysis of the Relationship Between Smoking and Premature Births

The scatter plot shows the connection between the number of **mothers smoking during pregnancy** and the number of **babies born prematurely (before 37 weeks)**. The trendline and equation help us understand how strong this relationship is.

The equation $y = 0.6623x - 8.67$ states that for every additional woman who smokes, the number of premature infants rises by around **0.66**. This is a substantial positive link between smoking and preterm births. The intercept (**-8.67**) seems unrealistic, yet it is part of how the model works.

The **R² value of 0.85** indicates that smoking accounts for **85%** of the changes in preterm births. This is a substantial association, indicating that smoking is most certainly a contributing factor to premature deliveries. However, other factors like maternal health, stress, and diet may also have an impact.

Looking at the graph, the majority of the data points coincide with the trendline, demonstrating consistency. There is one **outlier (orange)** in which smoking rates are high yet premature births are somewhat higher than predicted.

Conclusion

This analysis suggests that **reducing smoking among pregnant women could help lower the number of premature births**. While smoking is a key factor, other health and lifestyle aspects should also be considered when addressing premature birth rates.

Part B: Simulation Business Analytics

Question 1

Intervals of random numbers			
Time between Patient Arrival	Probability	Lower Limit	Upper limit
1	0.2	0	0.2
2	0.4	0.2	0.6
3	0.3	0.6	0.9
4	0.1	0.9	1

Question 2

2A)	Time between patient arrival	Probability	Lower Limit	Upper limit	Time between patient arrival	Num of Patients	Random Numbers	Patient inter arrival time
3	1	0.2	0	0.2	1	1	0.3813	2
4	2	0.4	0.2	0.6	2	2	0.2159	2
5	3	0.3	0.6	0.9	3	3	0.6515	3
6	4	0.1	0.9	1	4	4	0.2425	2
7						5	0.9177	4
8						Total=		13

2B)	Triage Nurse Time (min)	Probability	Lower Limit	Upper limit	Triage Nurse Time (min)	Num of Patients	Random Numbers	Triage nurse service Time
	2	0.20	0	0.2	2	1	0.8166	10
	5	0.60	0.2	0.8	5	2	0.1033	2
	10	0.20	0.8	1	10	3	0.4680	5
						4	0.9518	10
						5	0.7870	5
						Total=		32

2C)	Doctor Service Time	Probability	Lower Limit	Upper limit	Doctor Service Time	Num of Patients	Random Numbers	Doctor Service Time
	13	0.15	0	0.2	13	1	0.1358	13
	15	0.55	0.2	0.7	15	2	0.6874	15
	17	0.30	0.7	1	17	3	0.3056	15
						4	0.1826	15
						5	0.9290	17
						Total=		75

Business Analytics

Question 3

Patient	Inter Arrival time	Arrival Time	Triage Nurse Time	Triage Start	Triage End	Doctor Time	Doctor Start	Doctor End	Triage Wait	Doctor Wait	Total Time
1	2	2	10	2	12	13	12	25	0	0	23
2	2	4	2	12	14	15	25	40	8	11	36
3	3	7	5	14	19	15	40	55	7	21	48
4	2	9	10	19	29	15	55	70	10	26	61
5	4	13	5	29	34	17	70	87	16	36	74

Under Question 3, we conducted a spreadsheet simulation to see how the first five patients proceeded through the Emergency Department (ED) under Scenario 1, which had only one triage nurse and one doctor. Patient arrival times were calculated using inter-arrival data from Question 1, whereas triage and doctor service durations were determined using random numbers from Question 2. The model captured critical timestamps for each patient, including arrival to triage, waiting for the doctor, consultation, and total time spent in the emergency department. With only one nurse and one doctor available, patients were frequently forced to wait if someone else was already being seen, resulting in delays that accumulated. The simulation clearly revealed that patients arriving later tended to have greater wait times, demonstrating how limited personnel and uncertain service times might affect overall flow. This scenario serves as a valuable comparison for measuring future staffing improvements.

Formulas used:

Triage Start =Previous Triage start + Nurse Time

Triage End=Triage Start-Triage Nurse Time

Doctor Start =max(Triage End, available time doctor)

Doctor End = Doctor Start + Doctor Time

Total Time (ED)= Doctor End - Arrival Time

The formulas listed above were used to calculate critical timestamps and durations for each patient in the Emergency Department (ED). This included the triage start, which was computed by combining the previous patient's triage start to the previous patient's nurse's service time. For the first patient, this was unnecessary because their triage start time coincided with their arrival time. The same method was used to determine triage end, doctor start, doctor end, and total time spent in the ED. These aided in mapping how each patient progressed through the system and how long they stayed at each stage.

Question 5

In Scenario 2, which comprises one nurse and one doctor, patients spend an average of 58 minutes in the system, 26 of which are spent waiting. The longest time recorded was 88 minutes, while the shortest was 28. The primary delay occurs during triage owing to the restricted availability of a single nurse, which impacts how soon patients may be sent to the doctor. Scenario 3 adds a second nurse, which reduces triage wait times by allowing patients to see whichever nurse is available next. However, because there is still only one doctor, that phase continues to hold things down, resulting in average times that are almost the same as in Scenario 2. Based on this, hiring a second doctor would assist balance the workload and shorten overall wait times. The findings clearly demonstrate that doctor availability is the primary obstacle. Pairing two nurses with two doctors might considerably increase patient flow, make better use of staff time, and result in a more efficient system in general.

References

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<https://www.gov.scot/publications/criteria-define-levels-neonatal-care-including-repatriation-within-nhs-scotland-framework-practice/>

Week 8 Tutorial: Forecasting+ time series forecasting, smoothing methods+forecasting accuracy

Week 9 Tutorial: Forecasting, regression analysis, trend projection, multiplicative method

Week 10 Tutorial: Neural Network Forecasting

Week 11 Tutorial: Monte Carlo Simulation

Week 12 Tutorial: Discrete Event Simulation Modelling